



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 31.07.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

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Course No. : PHA F 415

Course Title : Pathophysiology

Instructor-In-Charge : **Rajeev Taliyan**

Instructors : NA

Tutorial/Practical Instructors: Rajeev Taliyan

1: Course Description:

This subjects covers physiological changes due to injury, disorders and disease. Study of important classes of disorder and diseases, including the progress, symptoms and mechanisms of action, therapeutic uses diagnostic and toxic manifestations.

2:Scope and objective of course: This course is intended to impart the knowledge regarding the physiological and pathophysiological changes at cellular and molecular levels in various disorders/diseases.

3: Text book (TB): Pathology of disease-An introduction to clinical medicine. Stephen J Mcphee et al. 8th Edition, McGraw-Hill, New Delhi,3rd Edition,2018

4: Reference Book:1: Robbins and Cotran Pathological Basis of Disease: S.L Robbins and R.S Cotran and vinay Kumar ,abbas , N. Fausto, 7th Edition, Elsevier,2018

2: Pathophysiology made incredibly Easy by springhouse, Michal Shaw,Lippincott, willium and willkins publisher,7th edition, 2014

3: Pathophysiology: clinical concepts of disease process by sylvia andersen, 6th edition ,2002

5: Course Plan

Module Number	Lecture Session	Reference (TB) /RB	Learning Outcome
1 General Introduction	L 1: 1-2:. Introduction to Patho-		Understand



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and aspect of Pathophysiology, Causes of cell Injury	physiology, Cell injury	TB-Ch-1&RB-Ch-2	pathophysiology, its scope, identifying the cause of cell injury, respond of cell to various stimuli, adaption- Hypertrophy/Hyperplasia: Reversible and irreversible cell injury
	3-4: Impact of cell injury, adaption to stress, Injury		
	5-6: Type of cell Injury		
2. Pathophysiology of inflammation	L2: 1-2: Basic process and Mechanisms involved in the process of inflammation	RB: Ch-2	Understanding the process and mechanisms of inflammation.
	3-4: Alterations in vascular permeability and blood flow, mediators of inflammation, brief outline of the process of repair. Inflammatory/allergic -non communicable diseases	TB: 3/RB-2	Understanding the relationship between inflammation and repair process mechanism Asthma and COPD
	5: Anti-inflammatory agents	TB: Ch-24	Understanding anti-inflammatory mechanism, advantages and disadvantages
3. Introduction to Pain and Joint disorders.	L3: 1-2 Physiology of pain Mechanism of pain process	RB: Ch- 2	Understanding the mechanism of different types of pain:
	3-4: Pathophysiology, Symptoms and diagnosis of RA and Gout	TB-Ch-24	Pathophysiology of Joint pain-Rheumatoid arthritis and Gout and their diagnosis
	5: Management of RA and Gout	TB: Ch-24	
4. Introduction to Neoplasia and cancer	L4: 1-2: Pathophysiology:Mol. and biochemical basis of cancer.	TB: Ch-5 RB: Ch-7	Understanding the process of cell division, differentiation and uncontrolled growth of cell:
	3-4: Symptoms and molecular mechanisms of breast cancer		Molecular mech. involved in genesis of different cancers i.e breast, blood cancer and cervical etc



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5. Heart and pathophysiology of cardiac Disorders.	L5:1. Heart Physiology and basic pathological changes	RB-Ch-12 TB-11	Understanding heart physiology and molecular and cellular changes involved in Cardiac disorders:
	2-5. Pathophysiology of Hypertension, and myocardial infarction	TB: 11	
6. Pathophysiology of G.I.T diseases, metabolic disorders	Pathophysiology of Ulcer, IBD		Biochemical changes, symptoms, diagnosis and management of Ulcer and metabolic disorders
	Diabetes, Insulin resistance	T 10.6-10.9 RL.M6L2	
7. Introduction of CNS and CNS Disorders.	L7: 1-3 Pathological basis and molecular changes in Epilepsy, Parkinson's, Alzheimer's diseases and Psychological disorders	TB: Ch-7 RL.M7LI	Understanding the symptoms, Mol and cellular changes in different types of CNS disorders
8. Pathophysiology of diseases of microbes/Viral	L 8: 1-3. Basic pathological changes in Tuberculosis, Leprosies and their management	TB: Ch-7. RLM8.L1	Understanding the symptoms, and in Mol cellular of changes different types bacterial and viral infections the Understanding COVID 19 pandemic
	4-5. Basic pathological changes and general aspect of viral disease: STD, AIDS disease and COVID-19	TB: 12.4-12.9 TB: 13.1-13.7RLM8L2,3 Notes.	
9. Introduction of different type of blood group, Blood Disease	L 9: 1-2: Types of blood group, Pathophysiology of different blood diseases: RBC platelet and WBC disorder,	T 14.1-14.7 Notes	Understanding the symptoms, Mol & cellular changes in different types of blood diseases

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	28	08/10/26	CB
Comprehensive Examination	180 Min.	38	10/12/26	CB + OB
Continuous Assessment	-	34	-	CB + OB



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*Continuous assessments will be in terms of home assignment/ Quiz and Viva-Voce, class presentation

7. Chamber Consultation Hour: Saturday 12-1.00 pm

8. Make-up policy: Generally, make-up will be considered for regular students only (75% attendance in lecture classes). It is solely dependent on the “genuineness” of the circumstances. The make-up application should be personally given to instructor-in-charge.

9. Notices: Concerning this course will be displayed on Department of Pharmacy notice board/Nalanda or via email.

Instructor-in-charge
\ PHA F 415